Lab 2 – TrashTag Product Specification

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1 Introduction

Litter in outdoor recreational areas such as parks, rivers, lakes, and trails remains a persistent environmental and operational issue. Improper disposal of waste disrupts natural ecosystems, poses safety hazards, and creates financial burdens for municipalities responsible for cleanup. In many cases, trash accumulates in areas that are difficult to monitor, making identification and removal inefficient and resource-intensive. Existing cleanup processes often rely on manual discovery by volunteers or city workers, resulting in delayed responses and incomplete coverage of affected areas.

TrashTag is designed to address these inefficiencies by providing a centralized platform for reporting and coordinating cleanup efforts. The system can allow individuals to capture and upload geotagged images of trash, allowing cleanup organizations to view and respond to reports in real time. By integrating location-based reporting, interactive mapping, and task coordination, TrashTag improves communication between the public and cleanup organizations. In addition, the system incorporates a gamification mechanism that rewards participation, encouraging continued user engagement and contributing to more consistent environmental maintenance.

1.1 Purpose

TrashTag is a web-based reporting and cleanup coordination platform designed for individuals and organizations involved in environmental cleanup efforts. The system enables users to submit geotagged images of trash, view reported locations on an interactive map, and track cleanup progress in real time.

Key features for users include uploading photographs with automatically captured location data, viewing nearby reported trash, and receiving updates when cleanup activities occur. TrashTag also incorporates a gamification system that rewards users with points for participation, encouraging continued engagement.

Cleanup organizations are provided with tools to monitor reported trash locations, claim cleanup tasks, and update the status of completed work. These features improve coordination efficiency and resource allocation compared to traditional cleanup methods.

1.2 Scope

The TrashTag prototype is designed to improve the identification and coordination of trash cleanup efforts in outdoor environments. The system provides users with the ability to submit geotagged reports, view these reports on an interactive map, and track cleanup progress.

The prototype will include the core features such as photo-based trash reporting, automatic geolocation tagging, real-time map visualization, cleanup task assignment, and status updates. Users will also be able to receive participation points through a gamification system.

While the prototype demonstrates the primary functionality of the system, some features may be simplified or simulated. The system is not meant to replace existing municipal waste management systems, but rather to supplement them by improving reporting accuracy and coordination efficiency.

1.3 Definitions, Acronyms, and Abbreviations

EXIF (Exchangeable Image File Format): Metadata embedded within digital images that stores information such as the date, time, and GPS coordinates of the image capture.

Gamification: The incorporation of game-like mechanics such as rankings and achievement systems to motivate user participation and increase engagement in cleanup operations.

Geotagging: The addition of GPS coordinates to images that enable precise location identification where trash was reported.

Illegal Dumping: The unauthorized disposal of trash and other hazardous materials onto private and/or public property.

Live Real-Time Updates: Live data synchronization displaying trash reports to both users and cleanup organizations instantaneously.

Major Functional Component Diagram (MFCD): A diagram that illustrates the relationships between the major hardware, software, and system components within an application.

Metadata: Descriptive information within data files such as date, location, and timestamp data in photos to be submitted to TrashTag.

Representational State Transfer (REST): An architectural style for designing networked applications using standard HTTP methods to allow communication between client and server systems.

1.4 References

Environmental Protection Agency (EPA). (2023). *Sources of Aquatic Trash.* https://www.epa.gov/trash-free-waters/sources-aquatic-trash

Keep America Beautiful. (2020). *2020 National Litter Study.* https://kab.org/litter-study

Texas Parks and Wildlife Department. (2019). *San Marcos River Task Force Report.* https://tpwd.texas.gov/publications/nonpwdpubs/san_marcos_river_taskforce

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1.5 Overview

This product specification provides the system architecture, features, and capabilities of the TrashTag prototype. The following sections describe the system structure, functionality, user roles, and operational requirements needed for implementation.

2 General Description

The TrashTag prototype will be developed as a web-based application with a centralized database. The system will support both general users and cleanup organizations through a unified platform. Most of the major functionalities of the system will be included in the prototype, although some features may be simplified compared to the real-world product due to time or budget restrictions.

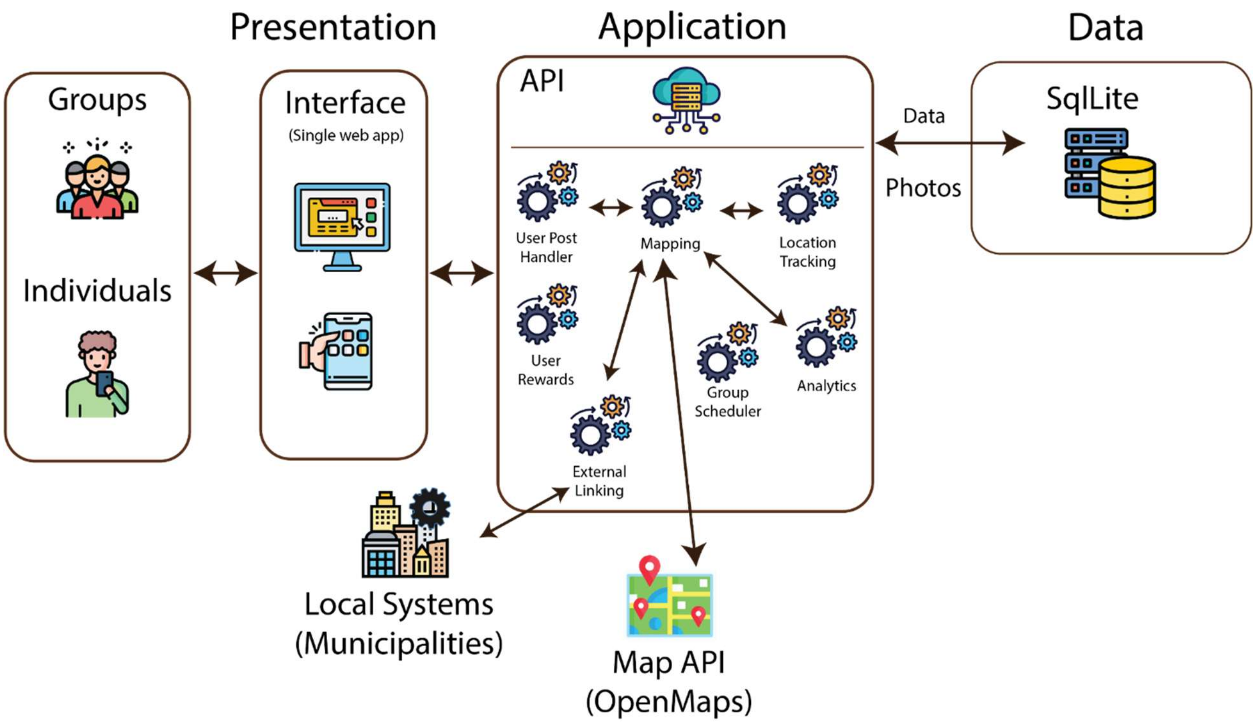
2.1 Prototype Architecture Design

Third-party services will be used to provide certain functionalities within the TrashTag prototype, such as map services that enable location-based reporting and visualization. The system will operate using a client-server architecture, where users interact with the platform through a web interface while backend services manage data processing and storage.

The architecture of the TrashTag system will consist of the presentation layer, the application layer, and the data layer. This design is illustrated in Figure 1.

Figure 1

TrashTag Major Functional Component Diagram



The presentation layer will include the frontend interface where users submit reports and view map data. The application layer will handle backend processing such as extracting geolocation data from images and updating report statuses. The data layer will manage storage of reports, user accounts, and cleanup activity records.

2.2 Prototype Functional Description

Most of the features of the real-world product will be implemented in the prototype, as can be seen in Table 1.

Table 1

TrashTag Feature Description and Prototype Implementation

Features	Description	Prototype Implementation
	Litter Reporting	
Report Litter Location	Users will pinpoint the location of the trash when reporting	Fully Implemented
Upload Photos	Users can provide one or more photo(s) of the trash they reported	Fully Implemented
Mark Trash Weight	Users can include an estimate of the trash's weight to help cleanup agencies come prepared	Fully Implemented
	Mapping	
View Litter Map	Users can see all reported litter nearby on an interactive map. They can also zoom out to see litter worldwide.	Fully Implemented
Litter Density Map	The litter map will indicate to users which areas have the most litter reports in red and will gradient to the areas with the least reports in yellow.	Eliminated – not an essential feature and, due to time constraints, will be difficult to implement.
	Cleanup Tracking & Verification	
Updated Cleanup Status	Reports will have their status updated if users confirm the report exists or removed if users confirm they no longer exist	Fully Implemented
Upload Cleanup Proof	Users who confirm a report has been cleaned up are required to provide proof of their claim (video, etc).	Eliminated – feature would require more cloud storage which would exceed out budget
	Community Impact & Analytics	
Gamification	Users are rewarded points for activity (reporting, cleaning, etc) and can compete. Exceptional users can earn cosmetic (non-monetary) rewards	Partially Implemented – points and leaderboard can be implemented but designed unique cosmetic rewards is too time consuming

Statistics	Allow cleanup groups and government agencies to download summary statistics so they can better allocate resources in their area.	Partially Implemented – only basic statistics will be gathered like number of reports and number of cleanups.

Users will be able to create reports by uploading images of trash, which will be processed to extract geolocation data and displayed on an interactive map. Cleanup organizations will be able to view reported locations, claim cleanup tasks, and update the status of reports once cleanup has been completed. The system will also include a gamification feature that allows users to earn points for reporting trash and participating in cleanup efforts. Some features may be partially implemented or simulated within the prototype, but the system will demonstrate the complete workflow from reporting to cleanup confirmation.

2.3 External Interfaces

The TrashTag prototype will use hardware, software, and user interfaces to provide functionality to the prototype's cross platform application.

2.3.1 Hardware Interfaces

The hardware required for the prototype is a computer with internet access with operating systems of either Mac, Windows, or Linux. A mouse and keyboard are also essential input systems for this prototype. Lastly, a separate or embedded camera is required for capturing images of trash.

2.3.2 Software Interfaces

The software required for the prototype includes React for frontend and Djnago/Python for backend. Digital ocean is used for cloud storage while SQLite is the database provider.

2.3.3 User Interfaces

To interact with the TrashTag prototype, the user will require a computer with internet access that has a keyboard and mouse, or a WIFI enabled mobile device capable of typing. They will also require a separate or embedded camera to take photos of encountered trash.

2.3.4 Communications Protocols and Interfaces

The application will use HTTP for communication through web browsers.